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Conservative Management of Urinary Incontinence

Behavioral and Pelvic Floor Therapy, Urethral and Pelvic Devices

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Keywords

bladder training; conservative behavioral treatments; lifestyle modification; pelvic floor muscle training; urinary incontinence

Questions

1. A person with cognitive impairment may not be a candidate for which behavioral interventions:
    a. prompted voiding
    b. timed voiding
    c. bladder training
    d. habit training
2. A bladder diary is integral to behavioral therapy because it can:
    a. help determine functional bladder capacity
    b. provide information about symptom improvement
    c. determine adherence to pelvic floor muscle training
    d. improve understanding of patient's voiding pattern

e. all of the above
f. b and d

3. Which of the following are true about pelvic floor muscle training?
    a. Pelvic floor muscle training is effective for treating stress, urge, or mixed incontinence in men and women of any age.
    b. Pelvic floor muscle training is appropriate for stress or mixed incontinence, but not for urgency incontinence.
    c. Pelvic floor muscle training works for women, but not for men.
    d. Pelvic floor muscle training is not effective for older people.
    e. Pelvic floor muscle training is useful for mild-moderate incontinence, but not severe incontinence.
4. When teaching the urge suppression technique, patients are encouraged to:
    a. Stay near a bathroom as much as possible, so they won't have far to go when they feel an urge to void.
    b. Stay away from the bathroom until they feel an urge to void. Then get to the bathroom as soon as possible.
    c. Stay away from the bathroom until urgency or the urge to void occurs. Then they should sit down, practice urge suppression techniques, and once the urge has passed, slowly walk to the bathroom to void.
    d. Wait as long as possible to go to the bathroom to increase bladder capacity.
5. When patients are trying to calm down urgency and prevent leakage, which of the following works BEST?
    a. Crossing the legs
    b. Squeezing the pelvic floor muscles while rushing to the bathroom
    c. Squeezing the pelvic floor muscles while sitting still until the urgency goes away, then walking to the bathroom
    d. Relaxing the pelvic floor muscles to help relax the bladder
6. When using fluid management as a treatment for urgency incontinence, patients are told to:
    a. consume a normal amount of fluid to avoid dehydration as well as sudden urgency.

b. increase fluid intake to ensure adequate hydration and train the bladder.
c. decrease fluid intake to minimize bladder filling.
d. increase fluid intake to prevent the loss of functional bladder capacity.

7. When attempting to identify dietary bladder irritants, the patient should:
    a. avoid spicy foods, tomatoes, and citric fruits
    b. eliminate caffeine
    c. keep a diary to see which foods or beverages increase urgency
    d. all of the above
8. When conducting pelvic floor muscle training, clinicians should:
    a. ensure that patients are contracting the pelvic floor muscles selectively
    b. prescribe a specific set of exercises for the patients to do each day
    c. teach patients to contract pelvic floor muscles whenever they engage in activities that precipitate leakage
    d. all of the above
9. Which of the following is true of caffeine reduction as a lifestyle modification for the treatment of incontinence?
    a. Patients should be told to eliminate caffeine gradually
    b. Patients should be encouraged to try eliminating caffeine for a few days to see how it affects their bladder
    c. Patients should be encouraged to reduce caffeine gradually
    d. Patients should be advised to eliminate coffee only, as it is the main source of caffeine

Answers

1. **c. Bladder training.** Bladder training is an example of "patient-dependent" interventions: it relies on active patient participation. There must be adequate function, learning capability, and motivation of the individual. Bladder training involves patient education regarding lower urinary tract function, setting

incremental voiding schedules, and teaching urge control techniques to help patients postpone voiding and adhere to the schedule.

2. **f. b and d.** A bladder diary provides information on the type and amount of fluid intake, type and frequency of symptoms, such as incontinence episodes, frequency of urination, the urgency associated with each, and the circumstances or reasons for incontinence episodes, which helps the provider plan appropriate components of behavioral intervention. A bladder diary is also the best noninvasive tool available to objectively monitor the patient's voiding habits and the effect of treatment on symptoms guiding the use of various treatment components.
3. **a. Pelvic floor muscle training is effective for treating stress, urge, or mixed incontinence in men and women of any age.** Behavioral interventions are well established for treating stress and urgency UI and OAB. Because behavioral treatments are effective and essentially risk free, they are the mainstay of conservative treatment and are recommended for both men and women of all ages as first-line therapy by several guidelines and consensus panels.
4. **c. Stay away from the bathroom until urgency or the urge to void occurs. Then they should sit down, practice urge suppression techniques, and once the urge has passed, slowly walk to the bathroom to void.** Patients are taught not to feel compelled to rush to the nearest bathroom when they feel the urge to void, believing that they are about to lose control. With behavioral training, they learn how this natural "gotta go" response is actually counter-productive, because it increases physical pressure on the bladder, increases the feeling of fullness, exacerbates urgency, and triggers detrusor contraction. Further, as the patient approaches the toilet, visual cues can trigger urgency and incontinence. To avoid this conditioned response, patients are taught not to rush to the bathroom when they feel the urge to void. Instead, they are advised to stay away from the bathroom, so as to avoid exposure to cues that trigger urgency. They are taught strategies to suppress urgency prior to walking to the toilet.

5. **c. Squeezing the pelvic floor muscles while sitting still until the urgency goes away, then walking to the bathroom.** Patients are taught not to rush to the bathroom when they feel the urge to void. Instead, they are advised to stay away from the bathroom, so as to avoid exposure to cues that trigger urgency. They are encouraged to pause, sit down if possible, relax the entire body, and contract PFM repeatedly, without relaxing in between contractions, to diminish urgency, inhibit detrusor contraction, and prevent urine loss. They focus on inhibiting the urge sensation, giving it time to pass. Once the sensation subsides, they walk at a normal pace to the toilet.
6. **a. Consume a normal amount of fluid to avoid dehydration as well as sudden urgency.** Fluid intake modifications depend on the patients' pattern of intake, which can be assessed by having them complete a 24- to 48-hour diary of intake and output, including voided volumes when possible. Reviewing such a diary can reveal excessive fluid intake, inadequate fluid intake, and diurnal patterns of intake that may be contributing to LUTS. Although it may seem counterintuitive, it is usually good advice to encourage patients to consume at least six 8-ounce glasses of fluid each day to maintain adequate hydration. Fluid intake should be regulated to 6- to 8-ounce glasses or 30 mL/kg body weight per day with a 1500 mL/day minimum at designated times unless contraindicated by a medical condition. The Institute of Medicine issued a report in 2004 with guidelines for total water intake for healthy people.
7. **c. Keep a diary to see which foods or beverages increase urgency.** A diary of food and beverage intake is useful for identifying which substances are in fact irritants for individual patients; and a trial period of eliminating these substances one at a time can be used to confirm the relationship.
8. **d. All of the above.** It is important to verify that patients have identified and can contract the pelvic floor muscles (PFMs) properly before initiating an exercise regimen. Specific exercise regimens vary considerably in frequency and intensity, and the ideal exercise regimen has not yet been determined. However, good results have been achieved in several trials using 45 to 60 paired contractions and relaxations per day. We use an "exercise

prescription" to prescribe the daily exercise program. It is important to teach patients how to prevent urine loss in daily life by occluding the urethra using active contraction of PFMs. Although exercise alone can improve urethral pressure and structural support and reduce incontinence, this motor skill enables patients to consciously occlude the urethra at specific times when urine loss is imminent. A careful history or examination of a bladder diary can alert the provider and patient of the circumstances during which each individual patient commonly experiences urine loss. Patients then learn to anticipate these activities and prevent leakage by contracting the PFM to occlude the urethra prior to and during coughing, sneezing, lifting, or any other physical activities that have precipitated urine leakage.

9. **b. Patients should be encouraged to try eliminating caffeine for a few days to see how it affects their bladder.** Many patients are reluctant initially to forgo their caffeinated beverages, but they may be convinced to try it for a short period of time, such as 3 to 5 days, to determine if they are sensitive to its effects. If they experience relief from their symptoms, they are often more willing to reduce or eliminate caffeine from their diet.

Chapter Review

1. Conservative behavioral treatments are a group of interventions that improve urinary incontinence and other lower urinary tract symptoms by changing patients' behavior or environment or by teaching new skills.
2. There is strong evidence that conservative interventions are effective first-line treatments for patients with urinary incontinence and other lower urinary tract symptoms.
3. The effectiveness of conservative treatments relies on the active participation of an involved and motivated patient.
4. Intensive behavioral interventions that involve an experienced clinician have been shown to be more effective.

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Electrical Stimulation and Neuromodulation in Storage and Emptying Failure

John P.F.A. Heesakkers, and Bertil Blok

Keywords

electrical stimulation; overactive bladder; percutaneous tibial nerve stimulation; sacral anterior root stimulation; sacral neuromodulation; transcutaneous electrostimulation; urge urinary incontinence

Questions

1. The current approved indications for sacral neuromodulation include all of the following EXCEPT:
    a. urinary urgency.
    b. urinary frequency.
    c. urgency urinary incontinence.
    d. interstitial cystitis.
    e. idiopathic nonobstructive urinary retention.
2. Which patient is NOT well suited for current neuromodulation therapies?
    a. A 65-year-old insulin-dependent diabetic man with bladder areflexia and nonobstructive urinary retention
    b. A 67-year-old woman who has had a cerebrovascular accident and now has urinary urgency and frequency

c. A 41-year-old woman with urgency urinary incontinence
d. A 55-year-old woman who has had vaginal sling surgery and urgency urinary incontinence
e. A 36-year-old woman with a history of interstitial cystitis with minimal pain who voids between 20 and 25 times per day

3. What reflex or reflexes are responsible for modulation of bladder function?
   a. Guarding reflex
   b. Bladder–Afferent loop reflex
   c. Bladder–Bladder reflex
   d. Bladder–Urethra reflex
   e. a and b
4. Which of the following is (are) considered the major clinical concern(s) associated with performing a sacral rhizotomy?
   a. Pelvic pain
   b. Creation of bladder areflexia
   c. Abnormal sexual function
   d. Pelvic and lower extremity sensory or motor abnormalities
   e. c and d
5. The S3 sensory and motor response pattern to electrical stimulation is best described as having which one of the following?
   a. Plantarflexion of the entire foot with sensation in the leg and buttock
   b. Levator reflex (bellows reflex) and sensations in the leg and buttock
   c. Dorsiflexion of the great toe and bellows reflex and pulling sensation in the rectum, scrotum, or vagina
   d. Plantarflexion of the first three toes of the foot and sensation of pulling in the rectum or vagina
   e. Bellows reflex (levator contraction) and sensation of pulling of the rectum
6. What is the main concern when performing magnetic resonance imaging (MRI) in the setting of neuromodulation and pacemaker-type devices?
   a. Potential of dislodgement of the pacemaker
   b. Heating of the electrical leads

c. Heating of the pacemaker
d. Potentially fatal arrhythmias
e. Significant neuromuscular injury risk

7. Which of the following represents the best clinical scenario for use of neuromodulation therapy in a patient with multiple sclerosis (MS)?
    a. Detrusor sphincter dyssynergy
    b. Bedridden with significant functional incontinence
    c. Mild symptoms with no potential need for future MRI
    d. A poorly compliant bladder
    e. Areflexic bladder
8. What skeletal landmarks are associated with the S3 nerve foramen?
    a. 9 cm from the tip of the coccyx
    b. 11 cm from the tip of the coccyx
    c. 13 cm from the tip of the coccyx
    d. The inferior aspect of the sacral iliac joints
    e. a and d
9. Perhaps the main reason why neuromodulation devices are not currently approved for use in the United States by the Food and Drug Administration in pediatric patients is due to:
    a. lack of efficacy.
    b. potential worsening of neuromuscular function due to bony abnormalities (spina bifida and myelomeningocele).
    c. lack of data on growth of the spinal cord and nerve roots in the setting of neuromodulation devices.
    d. worsening of bowel function (Hinman bladder syndrome).
    e. excellent results with noninvasive therapies (transcutaneous electrical nerve stimulation) and therefore no reason to perform more invasive sacral neuromodulation in the long term.
10. The best option for a patient who has undergone a failed stage I sacral neuromodulation for severe refractory urgency urinary incontinence is:
    a. anticholinergic therapy.
    b. B3 agonist therapy
    c. bilateral stimulation.

d. botulinum toxin A injections.
e. bladder augmentation.

11. An implantable pulse generator (IPG) infection would be best treated by:
    a. intravenous antibiotics.
    b. oral antibiotics.
    c. irrigation of the pocket.
    d. removal of the entire device.
    e. a and b.
12. Which of the following statements is FALSE about the Brindley device?
    a. It requires intact neuron pathways between the sacral cord and nuclei, pelvic nerve, and bladder to function.
    b. It works best in a state of long-term areflexic bladder function.
    c. It is used most often in patients with insufficient or nonreflex micturition after spinal cord injury.
    d. It is usually coupled with sacral posterior rhizotomy.
    e. Electrodes are applied to S2, S3, and S4 nerve roots.
13. Direct electrical stimulation of the bladder often results in all of the following EXCEPT:
    a. pelvic musculature contraction.
    b. erection.
    c. defecation.
    d. bladder neck opening.
    e. ejaculation.
14. Which one of the following statements regarding the use of the Brindley device is FALSE?
    a. It requires intact neural pathways between the sacral cord and the bladder.
    b. Sacral posterior rhizotomy is generally performed.
    c. Myogenic decompensation is a contraindication.
    d. Electrodes are applied extradurally to sacral roots S2 to S4.
    e. It utilizes the principle of poststimulation voiding.
15. Which of the following statements regarding neurostimulation or neuromodulation is FALSE?

a. The desired effect of neurostimulation is through direct stimulation of nerves and muscles.
b. Neurostimulation is mainly reserved for neurogenic conditions.
c. Neurostimulation produces a delayed clinical response.
d. The effect of neuromodulation is achieved through alteration of neurotransmission processes.
e. Neuromodulation may be useful for neurogenic as well as non-neurogenic conditions.

16. Which of the following studies is (are) the most useful in predicting which patients will or will not respond to sacral neuromodulation?
    a. Uroflow/postvoid residual monitoring
    b. Voiding diary
    c. Urodynamics/electromyography
    d. Percutaneous lead placement and trial stimulation
    e. c and d
17. Which of the following is (are) relative clinical contraindications for excluding potential candidates for neuromodulation and neurostimulation therapies?
    a. Patients with significant anatomic abnormalities in the spine or sacrum that may present challenges to gaining access
    b. Patients who cannot manage their devices or judge the clinical outcomes due to mental incapacitation
    c. Patients with physical limitations that prevent them from achieving normal pelvic organ function such as functional urinary incontinence
    d. Patients who are noncompliant
    e. All of the above
18. Which of the following statements best characterizes bilateral S3 nerve root stimulation for sacral neuromodulation therapy?
    a. It is a rational consideration for salvage therapy or added benefit as the bladder receives bilateral innervation.
    b. It is an approach alternative to failed unilateral stimulation in patients with urinary retention.
    c. Initial basis for this approach produced in spinal cord–injured animal models suggests this may be a potential approach in

humans.
d. All of the above.
e. a and c only.

19. Potential sites of selective nerve stimulation other than the S3 sacral root for neuromodulation therapies for pelvic health conditions include which of the following?
    a. S4 sacral root
    b. Pudendal nerve
    c. Dorsal genital nerve
    d. Posterior tibial nerve
    e. All of the above
20. When troubleshooting the complication of IPG site discomfort or pain, which of the following statements best describes the necessary action(s) needed?
    a. Rule out IPG site infection by physical examination.
    b. Turn off the device and ask the patient if the discomfort is still present to differentiate IPG pocket site issues from IPG electrical output-related causes.
    c. If the IPG discomfort is output related, check whether bipolar stimulation is better than unipolar stimulation.
    d. If IPG site discomfort is output related, check impedances because a current leak may be present from the neuroelectrode to extension lead connection.
    e. All of the above.
21. When patients report recurrent symptoms after reduction or improvement of symptoms with sacral neuromodulation therapy, which of the following should be undertaken to evaluate the reason for the loss of clinical efficacy?
    a. Check the device settings for inadvertent on/off changes and battery performance.
    b. Evaluate the stimulation perception and anatomic localization for changes.
    c. Check for intermittent stimulation perception via positional changes of the patient because this may suggest lead migration or a loose lead connection.

d. Obtain a radiograph to detect macro changes in the neuroelectrode position if findings in b and c are evident.
e. All of the above.

22. What is NOT true for intravesical electrostimulation (IVES)?
    a. It is mainly applied in patients with detrusor underactivity
    b. The mechanism is thought to come from direct efferent nerve stimulation
    c. The effect is higher in patients without underlying neurogenic disease
    d. Patients after hemorrhoidal surgery respond well to IVES.
23. What is true for percutaneous tibial nerve stimulation (PTNS)?
    a. It is one of the few treatments that have a proper sham-controlled study that underlines the efficacy.
    b. Evoked potential studies suggest that PTNS has no adaptive long-term modification in the brain.
    c. There might be some similarities of PTNS with SP5 activation of acupuncture.
    d. PTNS can induce toe flection of the contralateral foot.
    e. All of the above.
24. What is false for PTNS?
    a. There are good urodynamic data that give predictive indications for PTNS efficacy
    b. Efficacy of the known tibial implants is the same as the percutaneous technique.
    c. The PTNS implants have the same result as the SNS implants
    d. Battery life is critical for the PTNS devices
    e. All of the above

Answers

1. **d. Interstitial cystitis.** Although used commonly for interstitial cystitis (IC) symptoms, urgency/frequency IC is not truly an indication for the sacral neuromodulation devices. Several groups have seen benefits of sacral neuromodulation in IC patients, and there may be an expanding indication for this in the future.

2. **a. A 65-year-old insulin-dependent diabetic man with bladder areflexia and nonobstructive urinary retention. It is implied that the end organ response (bladder in this case) should have good function for sacral neuromodulation and, for that matter, any form of neuromodulation to work.** Neurostimulation may be different, but even if neurostimulation were used, simultaneous relaxation of the outlet would be required for a coordinated contraction and emptying phase to ensue.
3. **e. a and b. Two reflexes may play an important role in modulation of bladder function: the guarding reflex and the bladder afferent loop reflex. Both reflexes promote urine storage under sympathetic tone.** The guarding reflex guards or prevents urine loss in times of cough or other physical stress that would normally trigger a micturition episode. Suprapontine input from the brain turns off the guarding reflex during micturition to allow efficient and complete emptying. The bladder afferent reflex works through sacral interneurons that then activate storage through pudendal nerve efferent pathways directed toward the urethral sphincter. Similar to the guarding reflex, the bladder afferent reflex promotes continence during periods of bladder filling and is quiet during micturition.
4. **e. c and d. Bilateral anterior and posterior sacral rhizotomy or conusectomy converts a hyperreflexic bladder to an areflexic one. This alone may be inappropriate therapy because it also adversely affects the rectum, anal and urethral sphincters, sexual function, and the lower extremities.** In an attempt to leave sphincter and sexual function intact, selective motor nerve section was originally introduced as a treatment to increase bladder capacity by abolishing only the motor supply responsible for involuntary contractions.
5. **c. Dorsiflexion of the great toe and bellows reflex and pulling sensation in the rectum, scrotum, or vagina.** The characteristic response of the S3 nerve distribution based on its lower innervation is to the levator musculature (bellows contraction) of the anus and ipsilateral great toe contraction. The other answers suggest either S2 stimulation (leg rotation) or S4 levator contraction.

6. **b. Heating of the electrical leads.** Although many concerns exist for MRI and pacemaker devices, it has been shown that the main concern is heating of the electrical leads. This may, in turn, traumatize blood vessels, nerve roots, or other structures that the leads themselves are next to. Currently, MRI is contraindicated in the presence of a pacemaker.
7. **c. Mild symptoms with no potential need for future MRI.** It is unknown whether subcategories of MS patients (delayed emptying/storage dysfunction, areflexia, poor compliance) would be very good candidates for sacral neuromodulation, although it is doubtful based on disease severity alone. A mildly symptomatic patient without functional issues (e.g., can get to the bathroom in time with no major mobility issues) probably would be the best patient.
8. **e. a and d.** The measurements for the rough vicinity of the S3 nerve foramen have been tested by using the "cross hair" technique (Chai et al., 2000) [a] and simple measurements. The answers b and c are incorrect because they represent measurements from the anal verge (11 cm), and 13 cm is too far in general from the coccyx and would likely place one near S2 or S1.
9. **c. Lack of data on growth of the spinal cord and nerve roots in the setting of neuromodulation devices.** Pediatric patients have undergone sacral neuromodulation in off-label trials, but large-scale use has been limited by lack of data on the growth of the pediatric patient and the relation of the sacral lead with regard to the sacral nerve roots, and so on. Although noninvasive therapies have worked, they are limited by the need for continued repeat therapy to maintain durability of result.
10. **d. Botulinum toxin A injections.** The patient should have tried and failed anticholinergic or β 3 agonist therapy before having the sacral neuromodulation therapy. Scheepens and coworkers have shown that bilateral stimulation, although logical, has not been shown in urgency incontinent patients to lead to much improvement (Scheepens et al., 2002). One could argue that contralateral lead placement should be attempted, but no

prospective trials have shown that this makes a difference in outcomes. Because botulinum toxin A therapy has been proven to be as good as sacral neuromodulation, this is the most logical thing to do in this case.

11. **d. Removal of the entire device.** Because an IPG is a foreign body, it could harbor bacteria within a biofilm created by the infection. **Accordingly, it is best to have it removed in its entirety. Antibiotics and irrigation for the most part are temporizing measures. Furthermore, there is risk of an infection tracking along the sacral lead, which may create a sacral infection.**
12. **b. It works best in a state of long-term areflexic bladder function.** The chief applications of the Brindley device are in patients with inefficient or nonreflex micturition after spinal cord injury. Prerequisites for use are described by Fischer et al. (1993) as the following: (1) intact neural pathways between the sacral cord nuclei of the pelvic nerve and the bladder and (2) a bladder that is capable of contracting.
13. **d. Bladder neck opening. The spread of current to other pelvic structures the stimulus thresholds of which are lower than that of the bladder has often resulted in (1) abdominal, pelvic, and perineal pain; (2) a desire to defecate or defecation; (3) contraction of the pelvic and leg muscles; and (4) erection and ejaculation in males. It has also been noted that the increase in intravesical pressure was generally not coordinated with bladder neck opening or with pelvic floor relaxation and that other measures to accomplish voiding may be necessary.**
14. **d. Electrodes are applied extradurally to sacral roots S2 to S4.** Prerequisites for such use were described in one study as (1) intact neural pathways between the sacral cord nuclei of the pelvic nerve and the bladder and (2) a bladder that is capable of contracting. The chief application is in patients with inefficient or no reflex micturition after spinal cord injury. Simultaneous bladder and striated sphincter stimulation is obviated by sacral posterior rhizotomy, usually complete, which also (1) eliminates reflex incontinence and (2) improves low bladder compliance, if present. Electrodes are applied intradurally or extradurally to the S2, S3,

and S4 roots, but the pairs can be activated independently. The current Brindley stimulator uses the principle of *poststimulus voiding,* a term first introduced by Jonas and Tanagho (1975). Relaxation time of the striated sphincter after a stimulus train is shorter than the relaxation time of the detrusor smooth muscle. Therefore, when interrupted pulse trains instead of continuous stimulus trains are used, poststimulus voiding is achieved between the pulse trains because of the higher sustained intravesical pressure when compared with the striated sphincter.

15. **c. Neurostimulation produces a delayed clinical response.** In neurostimulation, the use of electrical stimuli on nerves and muscles has mainly been developed for achieving immediate clinical responses in neurogenic conditions of pelvic organ dysfunction, whereas in neuromodulation, the use of electrical stimuli to nerves has been developed for altering neurotransmission processes in cases of non-neurogenic as well as neurogenic conditions.
16. **d. Percutaneous lead placement and trial stimulation.** Despite all the studies done to date, there are no defined preclinical factors such as urodynamic findings that can predict which patients will or will not have a response to sacral neuromodulation. Thus, a trial of stimulation via a temporary or percutaneous lead placement is the best predictor of long-term clinical responsiveness.
17. **e. All of the above.** Whereas most patients who have failed more conservative therapies are considered candidates for neurostimulation and neuromodulation therapies, all of the above clinical considerations for excluding patients from this therapy should be considered. Furthermore, relative contraindications for patients who may be considering or who have an implantable electrical stimulation device are the issues of MRI and pregnancy.
18. **d. All of the above.** Bilateral stimulation has been suggested as an alternative, particularly in failed unilateral lead placements, for potential salvage or added benefit as the bladder receives bilateral innervation. The initial basis to consider bilateral stimulation was based on animal studies demonstrating that bilateral stimulation yielded a more profound effect on bladder inhibition than did

unilateral stimulation. Only one clinical study has been performed to demonstrate the potential differences in unilateral versus bilateral stimulation (Scheepens et al., 2002). This study showed no significant difference in outcomes for unilateral versus bilateral stimulation with regard to urgency urinary incontinence, frequency, or severity of leakage in the overactive bladder group, although, overall results were impressive in both categories. The patients in the retention group had better parameters of emptying (volume per void) in bilateral as compared with unilateral stimulation.

19. **e. All of the above.** The introduction of new stimulation methods as well as application of these methods to all the different nerve locations listed will continue to provide improved treatment alternatives, as shown in animal models and human applications. In addition, these innovations will provide the ability to further develop testable hypotheses of more basic questions on electrical neurostimulation, neuromodulation, and neurophysiology of the autonomic, somatic, and central pathways that regulate pelvic organ function.
20. **e. All of the above.** The probable causes of IPG site discomfort or pain are IPG pocket related or IPG output related. Pocket-related causes of discomfort include infection, pocket location (waistline), pocket dimension (too tight, too loose), seroma, and erosion. One should turn off the IPG and determine if the discomfort is still present to differentiate pocket-related from output-related cause. If the discomfort is persistent, the cause is not related to the IPG electrical output. In the absence of clinical signs of infection, IPG pocket-related causes such as pocket size, seroma, and erosion should be considered. If the discomfort disappears, the IPG electrical output is likely causing discomfort or pain. Output-related causes include sensitivity to unipolar stimulation if this mode is used or a current leak as demonstrated by abnormal impedances.
21. **e. All of the above.** When the patient presents with recurrent symptoms, one should evaluate the stimulation perception. The possibilities are that the patient perceives the stimulation in a wrong location as compared with baseline, has no stimulation, or

has intermittent stimulation based on lead migration or mechanical issues related to a loose connection or elevated impedances.

22. **b. The mechanism is thought to come from direct efferent nerve stimulation.** The main mechanism underlying IVES is thought to occur from direct field stimulation of Aδ afferents from low-threshold bladder mechanoreceptors, which comprise the sensory system responsible for initiating and maintaining the micturition reflex (Jiang et al., 1998). The other answers are all applicable for IVES.
23. a.**It is one of the few treatments that has a proper sham-controlled study that underlines the efficacy. This was shown by Peters et al. in 2010. Finazzi Agro showed in 2009 that evoked potential studies suggest that PTNS does have adaptive long-term modification in the brain via SS-LEP changes. There are similarities of PTNS with SP6 (Sanyinjiao) activation of acupuncture, not with SP5. PTNS can only induce toe flection of the ipsilateral foot.**
24. **e. All of the above. Although there are some urodynamic data for prediction of PTNS results (Vandoninck et al.), the overall availability of urodynamic data is scarce and contradictory. So we don't know the real value of this diagnostic tool for prediction of PTNS. There is no direct comparison yet of PTNS with a tibial nerve stimulation implanted device, although the expectation is that the implant is more efficacious. In line with this, there is no direct comparison of tibial nerve devices and sacral nerve devices. One of the assets of tibial nerve implants is that they are small because they lack a battery. Thus battery life is not an issue for the PTNS devices.**

Chapter Review

1. Electrical stimulation for lower urinary tract dysfunction is in its infancy compared to the standard of care in pain and cardiovascular disorders.

2. Nearly every nerve involved in the innervation of the lower urinary tract could theoretically be used to influence the lower urinary tract.
3. Sacral anterior root stimulation is very effective and improves quality of life dramatically but is also very invasive and irreversible.
4. Future innovation of the Brindley technique should avoid dorsal rhizotomy.
5. Of the peripheral types of neuromodulation, percutaneous tibial nerve stimulation is the best tested and developed one, and it has FDA approval.
6. New initiatives to have more patient- and doctor-friendly techniques are coming up.
7. Sacral neuromodulation is an established third treatment option for idiopathic bladder dysfunction. Sacral neuromodulation does have FDA approval for idiopathic overactive bladder and nonobstructive underactive bladder, but not for bladder or pelvic pain syndrome and neurogenic bladder.

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[a] Sources referenced can be found in *Campbell-Walsh-Wein Urology*, 12th Edition, on the Expert Consult website.

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Retropubic Suspension Surgery for Incontinence in Women

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Keywords

Burch colposuspension; mesh; retropubic repair; stress incontinence; urinary incontinence

Questions

1. Urodynamic stress urinary incontinence (SUI) refers to:
    a. incontinence that is demonstrated during a cough on clinical examination.
    b. incontinence occurring in the absence of urgency.
    c. incontinence occurring in combination with detrusor overactivity (DO).
    d. incontinence occurring on coughing in association with urgency and demonstrable DO.
    e. incontinence occurring on coughing in the absence of urgency and of urgency incontinence and with no demonstrable DO.
2. Anti-incontinence surgery via the retropubic route:
    a. is an effective approach for primary intrinsic sphincter deficiency (ISD).
    b. works by restoring the same mechanism of continence that was present before the onset of incontinence.

c. aims to improve the support to the urethrovesical junction and correct deficient urethral closure.
d. is the most effective form of anti-incontinence surgery.
e. is carried out laparoscopically as effectively as via an open approach.

3. The most important determinant affecting the outcome of retropubic surgery is:
   a. increasing age.
   b. postoperative activity.
   c. coexisting medical morbidity.
   d. previous surgery.
   e. obesity.
4. ISD is:
   a. present only in 30% of patients presenting with SUI.
   b. most likely present in the majority of women presenting with SUI.
   c. accurately identified on the basis of Valsalva leak point pressure.
   d. an absolute contraindication to a retropubic suspension procedure.
   e. clearly defined in the current literature.
5. Retropubic colposuspension procedures may act via which of the following mechanisms?
   a. Re-creating the normal continence mechanism
   b. Elevating the anterior vaginal wall and paravesical tissues toward the iliopectineal line
   c. Anchoring the obturator internus fascia to the iliopectineal line
   d. Suspending the bladder onto the periosteum of the symphysis pubis
   e. Strengthening the pubourethral ligaments
6. In assessing the outcome of retropubic suspension surgery, which of the following is most important?
   a. Using objective urodynamic-based outcome criteria
   b. Improving symptoms from the patient's perspective
   c. Achieving complete continence

d. Identifying the degree of improvement in the urethral closure pressure
e. Having follow-up data of at least 6 months' duration

7. Which of the following is not an indication for retropubic repair of SUI?
    a. A patient who needs a concomitant hysterectomy that cannot be performed vaginally
    b. A patient with urethral descent with straining and SUI
    c. A patient with limited vaginal access
    d. A patient who frequently generates high intra-abdominal pressure due to a chronic cough
    e. A patient with inadequate vaginal length or mobility of the vaginal tissues
8. Which of the following statements is TRUE regarding retropubic procedures for incontinence?
    a. It is important to avoid dissecting the old retropubic adhesions from prior incontinence procedures because these may contribute to continence.
    b. Nonabsorbable sutures are better than absorbable sutures for retropubic suspension procedures.
    c. It may be necessary to open the bladder to facilitate identification of the bladder margins and bladder neck.
    d. A urethral Foley catheter is preferred for bladder drainage because it is more comfortable and associated with fewer urinary tract infections and earlier resumption of voiding.
    e. The retropubic space must be drained after the procedure to prevent bleeding.
9. Which of the following statements is TRUE regarding the Marshall-Marchetti-Krantz (MMK) procedure?
    a. It is important to elevate the midurethra and external sphincter in particular.
    b. It carries little risk of causing urethral obstruction.
    c. It is associated with osteitis pubis.
    d. A better than 90% cure rate can be expected in the long term.
    e. The sutures should incorporate a full thickness of the vaginal wall and lateral urethral wall.

10. Which of the following is TRUE of the Burch colposuspension?
    a. It is appropriate only for patients with adequate vaginal mobility and capacity.
    b. The repair is performed between the vagina and the arcus tendineus fasciae pelvis bilaterally.
    c. It is less effective than a tension-free vaginal tape procedure.
    d. It is less effective than a paravaginal repair.
    e. It is more effectively performed via a vaginal approach.
11. Laparoscopic retropubic colposuspension is advantageous versus open colposuspension because:
    a. it is technically simple to perform.
    b. it provides access for repair of an associated central defect cystocele.
    c. it is more effective than an open colposuspension.
    d. it is associated with shorter hospitalization and recovery times.
    e. it is associated with shorter operating times.
12. Common complications specific to retropubic suspension procedures include:
    a. bladder denervation.
    b. detrusor sphincter dyssynergia.
    c. postoperative voiding difficulty.
    d. detrusor underactivity.
    e. genitourinary tract fistulae.
13. Postoperative voiding difficulty after a retropubic suspension procedure:
    a. is more likely if there is preexisting detrusor dysfunction.
    b. may be due to detrusor sphincter dyssynergia.
    c. is most likely to occur with undercorrection of the urethral axis.
    d. should be managed by ureterolysis within 1 month.
    e. occurs in less than 1% of patients.
14. Which of the following statements is TRUE regarding DO and retropubic suspension procedures?
    a. Preoperative DO is a contraindication to a retropubic suspension because it increases the risk of postoperative DO.

b. New-onset DO after a suspension procedure performed for SUI invariably resolves within 3 months.
c. DO occurs de novo, on average in fewer than 2% of the patients reported in the literature.
d. A history of voiding symptoms and new-onset storage symptoms as well as a retropubically angulated urethra usually suggests obstruction.
e. DO is not causally related.

15. Prolapse as a reported complication of retropubic repairs:
    a. is rarely associated with a central defect cystocele.
    b. results in genitourinary prolapse as a sequel to Burch colposuspension to occur in less than 10% of women.
    c. may aggravate posterior vaginal wall weakness, predisposing to enterocele.
    d. will be prevented by a synchronous hysterectomy.
    e. occurs only rarely after a paravaginal repair.
16. From comparative studies in the literature, which is correct about open retropubic colposuspension?
    a. It is not as effective as a pubovaginal sling.
    b. It is not effective in patients with a low leak point pressure.
    c. It is no more effective than an anterior colporrhaphy.
    d. It is less effective than a tension-free vaginal tape procedure.
    e. It is no more effective than a paravaginal repair.

Answers

1. **e. Incontinence occurring on coughing in the absence of urgency and of urgency incontinence and with no demonstrable DO**. SUI is the symptom of involuntary loss of urine during situations of increased intra-abdominal pressure such as coughing or sneezing. **The International Continence Society defines *urodynamic stress incontinence* as the involuntary loss of urine during increased intra-abdominal pressure during filling cystometry, in the absence of detrusor (bladder wall muscle) contraction** (Abrams et al., 2002). [a] Thus urodynamic evaluation is a prerequisite for the

diagnosis of urodynamic SUI. It is not clear, however, especially from the clinical management standpoint, whether a urodynamic diagnosis is imperative for successful treatment of SUI.

2. **c. Aims to improve the support to the urethrovesical junction and correct deficient urethral closure.** Surgical procedures to treat SUI generally aim to improve the support to the urethrovesical junction and correct deficient urethral closure. There is disagreement, however, regarding the precise mechanism by which continence is achieved in the "normal asymptomatic female" and therefore, not surprisingly, how restoration of "normality" is reestablished via surgical manipulation. Anti-incontinence surgery is generally used to address the failure of normal anatomic support of the bladder neck and proximal urethra and ISD. **Anti-incontinence surgery does not necessarily work by restoring the same mechanism of continence that was present before the onset of incontinence.** Rather, it works by a compensatory approach, creating a new mechanism of continence (Jarvis, 1994). The surgeon's preference, coexisting problems, and the anatomic features of the patient and her general health condition often influence the choice of procedure.

Current evidence would suggest that in adequately experienced hands, there is no difference in overall safety and efficacy between laparoscopic and open colposuspension. Clearly, another concern is how generalizable the data are on laparoscopic colposuspension because the majority of reported studies are from expert laparoscopists or surgeons working in specialized units. The evidence base on both laparoscopic and open colposuspension is limited by relatively short-term follow-up (robust data are needed out to 5 years) and the tendency toward small numbers, and poor methodology limits the interpretation of most studies with the exception of those reported by Carey and coworkers (2006) and Kitchener and colleagues (2006).

3. **d. Previous surgery.** Surgery for recurrent SUI has a lower success rate. One study has reported that **Burch colposuspension has an 81% success rate after one previous surgical procedure has**

failed, but this drops to 25% after two previous repairs and 0% after three previous operations (Petrou and Frank, 2001). Other series report excellent results for colposuspension performed after prior failed surgery. Maher and associates (1999) and Cardozo and colleagues (1999) have both shown good objective (72% and 79%) and subjective (89% and 80%) success rates with repeat colposuspension at a mean follow-up of 9 months. Nitahara and coworkers (1999) reported a 69% subjective success at a mean follow-up of 6.9 years.

The evidence on the duration of symptoms as a predictor of outcome is conflicting. **Age may not be a contraindication to colposuspension (with equivalent success rates in the elderly at long-term follow-up), although others reported less success with increasing age**. The influence of levels of postoperative activity has been inadequately studied, so no recommendations can be made. There is limited evidence that medical comorbidity may affect surgical outcomes, depending on the outcomes selected. Obesity as a confounding variable is the subject of conflicting evidence in the literature and has not been studied in a prospective fashion. **Approximately one-fourth of women undergoing urodynamic study have mixed urodynamic SUI and DO.** It is likely that the presence of concomitant DO lessens the success rate of surgery. There is no consensus in the literature as to whether the presence of ISD as assessed by urethral pressure profilometry has any influence in outcome of colposuspension.

4. **b. Most likely present in the majority of women presenting with SUI.** Hypermobility of the bladder neck and proximal urethra results from a weakening or loss of the supporting elements (ligaments, fasciae, and muscles), which in turn may result from aging, hormonal changes, childbirth, and prior surgery. **It seems likely that the majority of women with SUI will also have an element of intrinsic sphincteric weakness with a variable degree of loss of the normal anatomic support of the bladder neck and proximal urethra, resulting in hypermobility.**

A standardized test is not, however, available to differentiate the relative contributions of ISD and hypermobility, and therefore few studies have been able to accurately differentiate their individual contributions to the incontinence. Retropubic procedures act to restore the bladder neck and proximal urethra to a fixed, retropubic position and are used when hypermobility is thought to be an important factor in the development of that woman's SUI. This may facilitate the function of a marginally compromised intrinsic urethral sphincter mechanism, but if significant ISD is present, SUI will persist despite efficient surgical repositioning of the bladder neck and proximal urethra.

5. **b. Elevating the anterior vaginal wall and paravesical tissues toward the iliopectineal line.** Retropubic colposuspension urethral repositioning can be achieved by three distinctly different procedure principles. These are all based on a similar underlying principle but in a spectrum in relation to the degree of the support/elevations they achieve, and their outcomes differ somewhat in the longer term. **The Burch colposuspension is the elevation of the anterior vaginal wall and paravesical tissues toward the iliopectineal line of the pelvic side wall using two to four sutures on either side** (Burch, 1961). The vagino-obturator shelf repair aims to anchor the vagina to the obturator internus fascia and is a modification of a combination of the Burch colposuspension and paravaginal defect repair with placement of the sutures laterally anchored to the obturator internus fascia rather than hitching the vagina up to the iliopectineal line (Turner Warwick, 1986). The paravaginal defect repair aims to close a presumed fascial weakness laterally at the site of attachment of the pelvic fascia to obturator internus fascia (Richardson et al., 1976). The MMK procedure is the suspension of the vesicourethral junction (bladder neck) onto the periosteum of the symphysis pubis (Marshall et al., 1949). It aims to close the fascial defect rather than elevate the tissues in the paravesical area.
6. **b. Improving symptoms from the patient's perspective.** One or more high-quality validated symptom and quality-of-life instruments should be chosen at the outset of a clinical trial

representing the patient's viewpoint, accurately defining baseline symptoms, as well as any other areas in which treatment may be beneficial, and assessing the objective severity and subjective impact of bother. Although many, including the author, believe that urodynamic studies are helpful in defining the underlying pathophysiology in cases with incontinence, these tests have not been proven to have adequate sensitivity, specificity, or predictive value (Chapple et al., 2005). The International Consensus Meeting on Incontinence concluded that although urodynamic studies such as frequency-volume charts and pad tests were useful, there was inadequate evidence to justify pressure-flow studies for routine testing as either entry criteria or outcome measures in clinical trials, and they recommended that most large-scale clinical trials should enroll patients by carefully defined symptom-driven criteria when the treatment will be given on an empirical basis (Abrams et al., 2005).

7. **e. A patient with inadequate vaginal length or mobility of the vaginal tissues.** Although it has been suggested that a retropubic colposuspension should be considered in patients who frequently generate high intra-abdominal pressure (e.g., those with chronic cough from obstructive pulmonary disease and women in strenuous occupations), it has also been argued that these patients may be better served by a pubovaginal sling as well. **There may be specific indications for a retropubic approach for the correction of anatomic SUI, namely:**
   - **A patient undergoing a laparotomy for concomitant abdominal surgery that cannot be performed vaginally.**
   - **In cases with limited vaginal access.**

Conversely, contraindications include:

- **If there is a history of prior failed incontinence procedures, the existence of significant sphincteric deficiency must be suspected, even if hypermobility exists, and consideration given to performing a pubovaginal sling.**

- **In cases with a pan-pelvic floor weakness, a colposuspension should not be used in isolation but should be used as part of a comprehensive approach to the pelvic floor and be combined as appropriate with other alternative pelvic floor repair procedures.** Although lateral defect cystocele and enterocele lend themselves to retropubic repair, central defect cystocele, rectocele, and introital deficiency do not.
- **In cases in which there is an inadequate vaginal length or mobility of the vaginal tissues, such as after previous vaginal surgery or irradiation or after a previous vaginal incontinence procedure, a colposuspension should not be used.**
- **A retropubic colposuspension does not always adequately correct the associated vaginal prolapse that frequently coexists with bladder neck hypermobility.**

8. **c. It may be necessary to open the bladder to facilitate identification of the bladder margins and bladder neck.** In open retropubic suspension procedures, good access to the retropubic space is crucial. This is best performed with the patient in the supine position with the legs abducted, in either a low or a modified dorsal lithotomy position using stirrups, allowing access to the vagina during the procedure and a perineoabdominal progression. A urethral Foley catheter is inserted; the catheter balloon is used for subsequent identification of the urethra and bladder neck and is invaluable in allowing palpation of the edges of the bladder by appropriate manipulation. A Pfannenstiel or lower midline abdominal incision is made, separating the rectus muscles in the midline and sweeping the anterior peritoneal reflection off the bladder. It is essential to optimize the access to the retropubic space, and if a Pfannenstiel skin incision is made, it is advisable to utilize the suprapubic V modification described by Turner-Warwick and colleagues (1974). Likewise, whatever incision is made, extra valuable access to the retropubic space is obtained by extending the division of the rectus muscles right down to the pubic bone and elevating the aponeurotic insertion of the rectus muscle right off the upper border of the pubic bone.

The retropubic space is then developed by teasing away the retropubic fat and underlying retropubic veins from the back of the pubic bone. The bladder neck, anterior vaginal wall, and urethra are then easy to identify—often facilitated by the presence of the Foley catheter balloon. In patients who have had previous retropubic surgery, the dissection is performed sharply, and it is important to take down all old retropubic adhesions, particularly in the presence of a prior failed repair. **If difficulty is encountered in the identification of the bladder neck, the bladder may be partially filled or even opened to identify its limits; an examining finger in the vagina is invaluable in aiding the dissection** (Symmonds, 1972; Gleason et al., 1976).

It is important to identify the lateral limits of the bladder as it reflects off the vaginal wall, because only in this manner can one avoid inadvertent suturing of the bladder itself. Dissection over the bladder neck and urethra in the midline is to be avoided so as to not damage the intrinsic musculature. The lateral bladder wall may be "rolled off" medially and cephalad from the vaginal wall using a mounted swab and by using countertraction with a finger in the vagina. In the author's experience, it is necessary to incise the endopelvic fascia. Occasional venous bleeding from the large vaginal veins is controlled by suture ligature, although it often resolves with tying of elevating sutures. To aid in the identification of the lateral margin of the bladder, it is helpful to displace the balloon of the Foley catheter into the lateral recess, where it can be easily palpated through the bladder wall.

Absorbable sutures were used in the original descriptions of the MMK procedure (chromic catgut), Burch colposuspension (chromic catgut), and vagino-obturator shelf procedure (polyglycolic acid or polydioxanone), whereas the original paravaginal repair used nonabsorbable sutures (silicon-coated Dacron). Fibrosis during subsequent healing is likely to be the most important factor in providing continued fixation of the perivaginal fascia to the suspension sites (Tanagho, 1996); nevertheless, some surgeons believe that a nonabsorbable suture material is better because of the risk of suture dissolution before the development of adequate fibrosis (Penson and Raz, 1996). **Clearly the choice of suspension suture material is personal, but it must be remembered that nonabsorbent sutures eroding into the**

lumen of the bladder are a not uncommon complication and a not-uncommon source of medical litigation (Woo et al., 1995).

Some degree of immediate postoperative voiding difficulty can be expected after retropubic suspensions (Lose et al., 1987; Colombo et al., 1996). Immediately postoperatively, bladder drainage may take the form of a urethral or a suprapubic catheter, generally based on surgeon preference. A voiding trial is usually performed around the fifth day postoperatively. However, there is some evidence that a suprapubic catheter may be advantageous with respect to a lower incidence of asymptomatic and febrile urinary tract infection and earlier resumption of normal bladder function (Andersen et al., 1985; Bergman et al., 1987). In addition, the use of a suprapubic tube is generally more comfortable, allows the patient to participate in catheter management, and avoids the need for clean intermittent catheterization. Catheterization can be discontinued when efficient voiding has resumed, which is usually indicated by a postvoid residual volume either less than 100 mL or less than 30% of the functional bladder volume.

A tube drain may be placed in the retropubic space when there is concern about ongoing bleeding from perivaginal veins that may prove difficult to control with suture and electrocautery. Often, tying the suspension sutures is sufficient to stop this bleeding, but when it persists, drainage of the retropubic space is indicated. The drain is generally removed on the first to third day, when minimal output is noted.

9. **c. It is associated with osteitis pubis. Complications occur in up to 21% of cases (Mainprize, 1988), and the placement of sutures through the pubic symphysis incurs the risk of osteitis pubis, a potentially devastating complication of the MMK procedure that has been reported in 0.9% to 3.2% of patients** (Lee et al., 1979; Mainprize, 1988; Zorzos and Paterson, 1996). Patients usually present 1 to 8 weeks postoperatively with acute pubic pain radiating to the inner thighs and aggravated by moving. Physical examination reveals tenderness over the pubic symphysis, and radiography demonstrates haziness to the borders of the pubic symphysis and possibly lytic changes. Treatment is with bed rest, analgesics, and possibly corticosteroids (Lee et al., 1979).

10. **a. It is appropriate only for patients with adequate vaginal mobility and capacity. The Burch retropubic colposuspension, which has undergone few modifications since its original description, is appropriate only if the patient has adequate vaginal mobility and capacity to allow the lateral vaginal fornices to be elevated toward and approximated to the Cooper ligament on either side.**
11. **d. It is associated with shorter hospitalization and recovery times.** Proposed advantages to the laparoscopic approach include improved intraoperative visualization, less postoperative pain, shorter hospitalization, and quicker recovery times (Liu, 1993). Disadvantages include greater technical difficulty with resultant longer operating times and higher operating costs (Paraiso et al., 1999).

The last major publication in this field was a meta-analysis of all of the comparative studies published between 1995 and 2006 of laparoscopic versus open colposuspension (Tan et al., 2007). End points evaluated were operative outcomes and subjective/objective cure. A random-effect model was used and sensitivity analysis performed to account for bias in patient selection. Sixteen studies matched the selection criteria, reporting on 1807 patients, of whom 861 (47.6%) underwent laparoscopic and 946 (52.4%) underwent open colposuspension. Length of hospital stay and return to normal life were significantly reduced after laparoscopic surgery. These findings remained consistent on sensitivity analysis. Bladder injuries occurred more often in the laparoscopic group, but only with marginal statistical significance. Comparable bladder injury rates were found when studies were matched for quality, year, and randomized trials. Cure rates were similar between the two procedures at 2-year follow-up.

12. **c. Postoperative voiding difficulty.** As with any major abdominal or pelvic surgical procedure, intraoperative and perioperative complications that may occur after a retropubic suspension include bleeding, injury to genitourinary organs (bladder, urethra, ureter), pulmonary atelectasis and infection, wound infection or dehiscence, abscess formation, and venous thrombosis/embolism. **Other**

common complications more specific to retropubic suspension procedures include postoperative voiding difficulty, DO, and vaginal prolapse.

Nevertheless, the reported incidence of these problems is relatively low. In their meta-analysis, Leach and associates (1997) noted a 3% to 8% transfusion rate for retropubic suspensions and no significant difference in the overall medical and surgical complication rates among retropubic suspensions, needle suspensions, anterior colporrhaphy, and pubovaginal slings.

Ureteral obstruction has been reported rarely after Burch colposuspension, and it usually results from ureteral kinking after elevation of the vagina and bladder base, although direct suture ligation of the ureter can occur (Applegate et al., 1987). If identified intraoperatively, it is best remedied by removal of the offending ligature and temporary placement of a ureteral stent. The so-called *post-colposuspension syndrome*, which has been described as pain in one or both groins at the site of suspension, has been noted in as many as 12% of patients after a Burch colposuspension (Galloway et al., 1987). More recently, Demirci and associates (2001) reported the occurrence of groin or suprapubic pain in 15 of 220 women (6.8%) after Burch colposuspension with a follow-up of 4.5 years.

13. **a. Is more likely if there is preexisting detrusor dysfunction.** Postoperative voiding difficulty after any type of retropubic suspension is not uncommon, and undoubtedly its occurrence is more likely if there is preexisting detrusor dysfunction or denervation resulting from extensive perivesical dissection. In most cases, however, it is the result of overcorrection of the urethral axis, owing to sutures being inappropriately placed or excessively tightened. If they are placed too medially, sutures may also transfix the urethra or distort it.

Preoperatively, at-risk patients may be identified by their history of prior voiding dysfunction or episodes of urinary retention. Preoperatively, these women should be counseled carefully about the potential for postoperative voiding difficulty and the possible need for self-catheterization. Their

incontinence should be of sufficient magnitude that its correction offsets the risk of the need for self-catheterization.

Women with post-cystourethropexy voiding problems who have obstruction often do not exhibit the classic urodynamic features of obstruction. However, the history of postoperative voiding symptoms and associated new-onset bladder irritative symptoms and a finding of a retropubically angulated and fixed urethra generally indicate that obstruction does exist (Carr and Webster, 1997). In such cases, revision of the retropubic suspension by releasing the urethra into a more anatomic position resolves voiding symptoms in as many as 90% of cases (Webster and Kreder, 1990; Nitti and Raz, 1994; Carr and Webster, 1997).

The meta-analysis by Leach and coworkers (1997) noted that the risk of temporary urinary retention lasting more than 4 weeks postoperatively was 5% for all retropubic suspensions, the risk for permanent retention was estimated to be less than 5%, and these risks were not significantly different from those for needle suspensions or pubovaginal slings.

14. **d. A history of voiding symptoms and new-onset irritative symptoms as well as a retropubically angulated urethra usually suggests obstruction. Bladder hyperactivity commonly accompanies anatomic SUI, and its incidence preoperatively has been reported to be as high as 30% in patients undergoing either first correction or repeated operations** (McGuire, 1981). Provided that it is considered as a diagnosis, urodynamic study is performed to show whether DO is present, an attempt at treatment of the related overactive bladder symptoms has been made (with or without success), and the patient has been advised that the presence of DO will increase the risk of continuing storage symptoms postoperatively, then preoperative bladder overactivity does not contraindicate a retropubic suspension procedure, provided that anatomic SUI has also been demonstrated. **In the majority of cases the bladder overactivity symptoms resolve after surgical repair** (McGuire, 1988). Leach and coworkers' meta-analysis (1997) found the risk of urgency after a retropubic suspension was 66% if urgency and DO were present preoperatively, 36% if there

was urgency but no documented overactivity preoperatively, and only 11% if there was neither urgency nor overactivity preoperatively. There was no significant difference in the incidence of postoperative urgency among retropubic suspensions, needle suspensions, and pubovaginal slings. Postoperative urgency was noted in only 0.9% of MMK procedures in Mainprize and Drutz's meta-analysis of 15 series (1988), although Parnell and associates (1982) reported that 28.5% of their patients developed postoperative storage symptoms. Jarvis's meta-analysis (1994) of Burch colposuspensions found the incidence of de novo bladder overactivity to be 3.4% to 18%. More recently Smith and associates quote a figure for postoperative DO of 6.6% for colposuspension (range 1.0% to 16.6%), whereas the incidence of postoperative urgency or urgency incontinence after the paravaginal/vagino-obturator shelf repair has been reported to be 0% to 6% (Shull and Baden, 1989; German et al., 1994; Colombo et al., 1996).

For those patients in whom postoperative storage symptoms persist, proven to be associated with DO and intractable to management with anticholinergic therapy and behavioral modification, surgical techniques including intravesical botulinum toxin therapy, neuromodulation, augmentation cystoplasty, or detrusor myectomy may be indicated.

Bladder storage symptoms arising de novo after retropubic suspension may be associated with bladder outlet obstruction. This premise is supported by the frequent coexistence of these symptoms with impaired voiding after suspension procedures and confirmed by the finding that ureterolysis, by freeing the urethra from an obstructed position, often resolves both storage and voiding symptoms (Raz, 1981; Webster and Kreder, 1990).

15. **c. May aggravate posterior vaginal wall weakness, predisposing to enterocele.** Retropubic suspensions alter vaginal and bladder base anatomy, and, thus, postoperative vaginal prolapse is a potential complication. **Genitourinary prolapse has been reported as a sequel to Burch colposuspension in 22.1% of women (range 9.5% to 38.2%) by Smith and colleagues (2005)**

in their review of the literature. The Burch colposuspension, because of lateral vaginal elevation, may aggravate posterior vaginal wall weakness, predisposing to enterocele. The incidence varies between 3% and 17% (Burch, 1961, 1968; Galloway et al., 1987; Wiskind et al., 1992), and because of this, prophylactic obliteration of the cul-de-sac of Douglas is sometimes considered when performing retropubic suspensions (Shull and Baden, 1989; Turner-Warwick and Kirby, 1993). **However, simultaneous hysterectomy is not recommended prophylactically because it does not enhance the outcome of a retropubic suspension and should be performed only if there is concomitant uterine pathology** (Milani et al., 1985; Langer et al., 1988). Although the Burch colposuspension and paravaginal/vagino-obturator shelf repair both correct lateral defect cystourethroceles, recurrent cystourethroceles were noted in 11% and 39% of Burch colposuspensions and paravaginal repairs, respectively (Colombo et al., 1996). In Mainprize and Drutz's review (1988), postoperative cystocele was noted in only 0.4% of patients after an MMK procedure.

Wiskind and coworkers (1992) noted that 27% of patients who had undergone a Burch colposuspension developed prolapse requiring surgery: rectocele in 22%, enterocele in 11%, uterine prolapse in 13%, and cystocele in 2%. More recently, it has been suggested that most women are asymptomatic and fewer than 5% have been reported to request further surgery (Smith et al., 2005). Ward and associates (2004) reported 4.8% of women needing a posterior repair whereas Kwon and coworkers (2003) reported 4.7% requiring subsequent pelvic reconstruction.

Because retropubic suspensions are unable to correct central defect cystoceles, patients must be carefully examined preoperatively to exclude their presence.

16. **a. It is not as effective as a pubovaginal sling.** A total of 655 women were randomly assigned to study groups: 326 to undergo the sling procedure and 329 to undergo the Burch colposuspension; 520 women (79%) completed the outcome assessment (Aldo et al.,

2007). At 24 months, success rates were higher for women who underwent the sling procedure than for those who underwent the Burch colposuspension for both the overall category of success (47% vs. 38%, $P = .01$) and the category specific to SUI (66% vs. 49%, $P < .001$). There was no significant difference between the sling and Burch colposuspension groups in the percentage of patients who had serious adverse events (13% and 10%, respectively; $P = .20$). However, more women who underwent the sling procedure had adverse events than in the Burch colposuspension group, with 415 events among 206 women in the sling group as compared with 305 events among 156 women in the Burch colposuspension group. This difference was due primarily to urinary tract infections: 157 women in the sling group (48%) had 305 events and 105 women in the Burch colposuspension group (32%) had 203 events. When urinary tract infections were excluded, although the rates of adverse events were similar in the two groups, there was more difficulty voiding. The distribution of time to return to normal voiding differed significantly between the two groups: Voiding dysfunction was more common in the sling group than in the Burch colposuspension group (14% vs. 2%, $P < .001$). Consequently, surgical procedures to reduce voiding symptoms or improve urinary retention were performed exclusively in the sling group, in which 19 patients underwent 20 such procedures (63% vs. 47%, $P < .001$). Treatment satisfaction rates for the 480 subjects who answered the satisfaction question at 24 months were significantly higher in the sling group than in the Burch colposuspension group (86% vs. 78%, $P = .02$). A further analysis of this study focused on sexual activity as assessed by the Pelvic Organ Prolapse/Urinary Incontinence Sexual Questionnaire (PISQ-12) among those sexually active at baseline and 2 years after surgery (Brubaker et al., 2009). This report demonstrated that sexual function improves after successful surgery and does not differ between Burch colposuspension and sling procedures.

It can therefore be reliably concluded that in specialist centers working in a standardized fashion, the autologous fascial sling results in a higher rate

of successful treatment of SUI but also greater morbidity than the Burch colposuspension.

Comparisons between the MMK and the Burch colposuspension procedures have generally yielded similar results. Three articles that reviewed the literature on incontinence procedures all found retropubic suspensions to be more effective than either needle suspensions or anterior colporrhaphies (Jarvis, 1994; Black and Downs, 1996; Leach et al., 1997). Most studies in the literature have not demonstrated a significant difference in cure rates between retropubic suspensions (generally a Burch colposuspension) and pubovaginal slings (Jarvis, 1994; Black and Downs, 1996; Leach et al., 1997). The literature on the paravaginal repair is sparse. The only randomized study that compared the Burch colposuspension with a paravaginal repair found significantly greater subjective and objective cure with the Burch colposuspension (Colombo et al., 1996). At this point, the tension-free vaginal tape procedure appears to be at least equivalent to the Burch colposuspension.

Chapter Review

1. Anti-incontinence surgery does not work by restoring the normal mechanism of continence but, rather, by a compensatory approach creating a new mechanism of continence.
2. ISD is suggested by a leak point pressure less than 60 cm H_2O or a maximum urethral closure pressure of less than 20 cm H_2O.
3. Approximately 40% of nulliparous 30- to 49-year-old women experience some degree of incontinence with exercise.
4. If nonabsorbable sutures are used in a retropubic suspension, they may migrate into the bladder and serve as a foreign body nidus for stone formation and infection.
5. The postoperative risk of SUI in continent women undergoing an abdominal sacrocolpopexy is substantially reduced by the addition of a Burch colposuspension.
6. A maximum urethral closing pressure of less than 20 cm H_2O is a contraindication to the Burch colposuspension.

7. All patients, before any colposuspension, should be advised about the potential need for intermittent self-catheterization.
8. The Burch colposuspension is regarded as the standard open retropubic procedure for incontinence.
9. If ISD is the primary problem, a fascial sling procedure should be performed rather than a colposuspension.
10. A MMK procedure, a paravaginal defect repair, and needle suspension procedures are not recommended for the treatment of SUI.
11. The majority of women with SUI will also have an element of intrinsic sphincteric weakness with a variable degree of loss of the normal anatomic support of the bladder neck and proximal urethra, resulting in hypermobility.
12. Contraindications for a retropubic colposuspension include significant sphincteric deficiency, pan-pelvic floor weakness, and inadequate vaginal length or mobility. Moreover, it may not correct the associated vaginal prolapse.
13. Bladder hyperactivity commonly accompanies anatomic SUI, and its incidence preoperatively has been reported to be as high as 30%. In the majority of cases, the bladder overactivity symptoms resolve after surgical repair.
14. Simultaneous hysterectomy is not recommended prophylactically because it does not enhance the outcome of a retropubic suspension and should be performed only if there is concomitant uterine pathology.

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[a] Sources referenced can be found in *Campbell-Walsh-Wein Urology, 12th Edition*, on the Expert Consult website.

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Vaginal and Abdominal Reconstructive Surgery for Pelvic Organ Prolapse

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Keywords

pelvic organ prolapse; reconstructive surgery

Questions

1. Which of the following statements regarding anterior colporrhaphy is FALSE?
    a. Anterior colporrhaphy is not used to treat stress incontinence.
    b. Recent series report a 40% recurrence rate for standard anterior colporrhaphy.
    c. The most likely contributing factor for failure of anterior compartment repairs is the concomitant presence of other compartmental defects.
    d. Cystoscopy with indigo carmine is not routinely necessary.
    e. Ensuring that the bladder is drained before perforating the endopelvic fascia may decrease bladder injuries.
2. A 50-year-old woman presents with symptoms of voiding difficulty and vaginal bulging. She has had a prior hysterectomy. In the supine position she demonstrates stage 3 prolapse with the Ba point at +4. Her C point is −4 with a TVL of 9 cm, and her Bp is −3.

With prolapse reduction there is no stress incontinence. Which of the following repairs would be most appropriate?

a. Vault suspension alone
b. Vault suspension with anterior repair
c. Vault suspension with anterior repair and anti-incontinence procedure
d. Vault suspension with anterior repair, anti-incontinence procedure, and posterior repair
e. Anterior repair alone
f. Anterior and posterior repair

3. Which surgical technique of rectocele repair is most associated with postoperative dyspareunia?
    a. Levator plication
    b. Site-specific repair
    c. Site-specific repair with biologic interposition graft
    d. Transanal repair of rectocele
    e. Perineorrhaphy
4. Which symptom changes the least following site-specific posterior colporrhaphy?
    a. Dyspareunia
    b. Constipation
    c. Vaginal mass
    d. Splinting
    e. Vaginal pressure or pain
5. When planning a multi-compartment repair with an outlet procedure, which of the following is the preferred sequence of compartmental approaches?
    a. Apical, anterior, posterior, sling
    b. Sling, apical, anterior, posterior
    c. Apical, anterior, sling, posterior
    d. Posterior, anterior, apical, sling
    e. Anterior, posterior, sling, apical
6. When performing a posterior repair, the literature currently supports which of the following methods to maximize success rates and decrease complications, namely dyspareunia?
    a. Traditional posterior colporrhaphy

b. Mesh augmented posterior colporrhaphy
c. Traditional posterior colporrhaphy with levator plication
d. Porcine graft augmented posterior colporrhaphy
e. Cadaveric graft augmented posterior colporrhaphy

7. Which of the following procedures have proven to be ineffective in the treatment of stress urinary incontinence?
   a. Synthetic midurethral sling
   b. Anterior colporrhaphy
   c. Burch colposuspension
   d. Autologous pubovaginal sling
   e. Vaginal paravaginal repair
   f. b and e
   g. All of the above
8. Which of the following statements regarding anterior compartment prolapse and/or repair is false?
   a. An anterior compartment defect may be central, lateral, or combined.
   b. A central defect results from midline separation or attenuation of the pubocervical (perivesical) fascia.
   c. A lateral defect results when the pubocervical fascia detaches from the arcus tendineus fascia pelvis (ATFP).
   d. Most anterior compartment defects have concomitant lateral and/or apical defects.
   e. Isolated anterior colporrhaphy is not recommended when additional defects are present.
   f. Anterior compartment prolapse repair also suspends the vaginal apex.
9. Which of the following defects requires surgical correction?
   a. Ba −1 in an asymptomatic 48-year-old
   b. Bp −3 with complaints of constipation
   c. C −9, TVL 10, Ba −3
   d. Ba 0 with complaint of a vaginal bulge
   e. C −8, TVL 10, Ba −2
10. A 31-year-old woman with a history of four prior vaginal deliveries presents to your office with symptoms of a vaginal bulge. On pelvic exam, you note the cervix is protruding 4 cm beyond the

introitus, the anterior vaginal wall bulges just beyond the hymenal ring, and the posterior wall seems to be well supported. Total vaginal length is about 9 cm. The patient is not interested in using a pessary. She desires future fertility. She wants you to do the operation that will reduce her chance of requiring another surgery for prolapse in the future.

The next best step in management is:

a. Vaginal hysterectomy and uterosacral ligament suspension
b. Vaginal hysterectomy and sacrospinous ligament suspension
c. Transvaginal suture hysteropexy
d. Abdominal sacrohysteropexy with mesh
e. Abdominal hysterectomy and sacrocolpopexy

11. A 48-year-old active woman with incomplete uterovaginal prolapse with cervical elongation desires a transvaginal uterine-sparing procedure. POQ point C is +2 and Point D is −7. Select the most appropriate type of uterine-preserving POP reparative surgery for her condition.
    a. Le Forte colpocleisis
    b. Sacrohysteropexy
    c. Manchester procedure
    d. Sacrospinous cervicopexy
    e. Uterosacral hysteropexy

12. An 89-year-old woman who has failed numerous pessaries has symptomatic complete uterovaginal prolapse. She is frail, lives in a nursing home, and has medical comorbidities making her high risk for general anesthesia. Select the most appropriate surgical procedure.
    a. Uterosacral hysteropexy
    b. Sacrospinous cervicopexy
    c. Sacrohysteropexy
    d. Le Forte colpocleisis
    e. Manchester-Fothergill operation

13. A 54-year-old woman who has a history of multiple laparotomies for treatment of endometriosis and a history of small bowel obstruction requiring resection and anastomosis presents with symptomatic incomplete uterovaginal prolapse. POPQ C is at 1, Point D is −4. During her small bowel surgery, the surgeon noted significant pelvic adhesions involving the uterus, tubes, and ovaries, which were not addressed during that surgery. Select the most appropriate surgical procedure.
    a. Uterosacral hysteropexy
    b. Sacrospinous cervicopexy
    c. Sacrohysteropexy
    d. Le Forte Colpocleisis
    e. Manchester-Fothergill operation
14. A 41-year-old woman with symptomatic uterovaginal prolapse with POPQ point C +2, D −6 presents to your office desiring a uterine-sparing procedure. She smokes a half a pack of cigarettes per day and will not quit. She also has a history of two procedures to treat recurrent mid-urethral polypropylene sling exposure. She also has a 5-cm ovarian cyst that appears benign but has been present for 8 months; she desires its removal. She is sexually active and is concerned about pain with intercourse after the procedure. Select the most appropriate surgical procedure.
    a. Uterosacral hysteropexy
    b. Sacrospinous cervicopexy
    c. Sacrohysteropexy
    d. Le Forte colpocleisis
    e. Manchester-Fothergill operation
15. A 28-year-old woman who has Ehlers-Danlos syndrome presents with symptomatic uterovaginal prolapse. Her relevant POPQ is as follows: C +1, TVL 8, D −4. She desires future fertility. She has a history of asthma and problems with chronic constipation. Select the most appropriate procedure for this patient.
    a. Uterosacral hysteropexy
    b. Sacrospinous cervicopexy
    c. Sacrohysteropexy
    d. Le Forte colpocleisis

e. Manchester-Fothergill operation

16. A 60-year-old patient undergoes a transvaginal hysterectomy (TVH), bilateral salpingo-oophorectomy (BSO), uterosacral ligament suspension, enterocele repair, anterior and posterior repair, and transvaginal tape (TVT) sling procedure for symptomatic uterovaginal prolapse (POPQ point C +3, D-5) and stress urinary incontinence. She presents 6 months postoperative and tells you she is having pain with intercourse. Select the procedure that is the probable cause of her dyspareunia.
    a. Vaginal vault suspension
    b. Enterocele repair
    c. TVT sling
    d. Anterior colporrhaphy
    e. Posterior colporrhaphy
17. A 46-year-old woman with vaginal vault prolapse undergoes a laparoscopic sacrocolpopexy for symptomatic vaginal vault prolapse. Her BMI is 19 and she has a history of lupus. The patient is positioned by a new nurse circulator. The surgery lasts 5 hours. When seeing her on POD#1, she is unable to flex her knee, has foot drop, and decreased sensation in the calf. Select the likely nerve affected.
    a. Femoral
    b. Genitofemoral
    c. Sciatic
    d. Brachial plexus
    e. Obturator
    f. Common peroneal
    g. Ilioinguinal
    h. Iliohypogastric
18. A 57-year-old woman with symptomatic uterovaginal prolapse presents desiring surgical correction of her prolapse. She has no significant comorbidities other than well-controlled hypertension. She is sexually active. She has been menopausal for the last 6 years and reports no postmenopausal bleeding. Her last pap smear was negative for malignancy; she has never had an abnormal pap smear. On manual exam her uterus is small and mobile. Her POPQ is as

follows: Aa +3, Ba+5, C +7, Gh 5, Pb 1, TVL 10, Ap −1, Bp −1, D+4. Select the best procedure to address her prolapse.

a. Anterior and posterior colporrhaphy
b. Vaginal hysterectomy with an anterior and posterior colporrhaphy
c. Abdominal supracervical hysterectomy with sacrocervicopexy ± anterior and posterior colporrhaphy
d. Abdominal hysterectomy
e. Lefort colpocleisis

19. Which of the procedures listed below to address pelvic organ prolapse has been reported to have the highest risk of ureteral injury?

a. Sacrospinous ligament suspension
b. Uterosacral ligament suspension
c. Anterior colporrhaphy
d. Transvaginal hysterectomy
e. Posterior colporrhaphy

20. A 51-year-old woman presents 2 months after a laparoscopic sacrocolpopexy with persistent low back pain. She states the pain has been present for the last 3 weeks and it does not feel like it is getting better. She describes the pain as constant and achy. Over-the-counter ibuprofen helps, but the pain never goes away. She admits to intermittent fever and chills but denies abdominal pain, nausea, or vomiting. She admits to having regular bowel movements and denies dysuria. Select the most likely diagnosis.

a. Herniated disc
b. Bowel obstruction
c. Osteomyelitis
d. Spinal stenosis

21. You are instructing your first-year fellow through a sacrospinous ligament suspension and tell your fellow that, to avoid complications related to suture placement, the suture should be placed:

a. 1 cm lateral and slightly superior to the ischial spine
b. 3 cm medial to the ischial spine, slightly below the superior edge of the ligament

c. 3 cm medial to the ischial spine, superiorly and slightly behind the ligament
d. 1 cm medial to the sacrum, slightly below the superior edge of the ligament

22. An 80-year-old woman has recurrent vaginal vault prolapse. She had a sacrospinous ligament fixation 20 years prior. She has not been sexually active for the last 15 years. She has failed numerous pessary trials. She is interested in a surgical procedure with the lowest likelihood for repeat surgery. Select the best procedure.
    a. Mesh sacrocolpopexy
    b. Vaginal mesh for prolapse
    c. Sacrospinous ligament fixation
    d. Colpocleisis

Answers

1. **d. Cystoscopy with indigo carmine is not routinely necessary.** Cystoscopy after the administration of indigo carmine or methylene blue is recommended as a routine practice after anterior colporrhaphy. If blue-tinged urine is not seen effluxing from each ureteral orifice, catheterization may be considered before taking down the plication sutures. Appropriate steps must be performed to ensure ureteral patency, including takedown of the sutures.
2. **b. Vault suspension and anterior repair.** The patient has stage 3 anterior vaginal wall prolapse at the apical descent. These defects are best addressed by restoring the apical and anterior support. Failure rates for an anterior repair alone are unacceptably high. As no stress urinalysis incontinence was demonstrated, there is no absolute indication for an anti-incontinence procedure. The support of the posterior compartment is intact and posterior repair is not indicated.
3. **a. Levator plication.** Levator plication was associated with high rates of de novo postoperative dyspareunia. As such, it has largely been abandoned.

4. **b. Constipation.** Constipation may be associated with dysmotility disorders of the rectum that do not improve with an anatomic repair. The symptoms most likely to improve are dyspareunia, the need for vaginal splinting, vaginal mass, and pressure.
5. **c. Apical, anterior, sling, posterior.** The apical compartment should always be addressed first. The anterior compartment should preferentially be repaired before the posterior compartment when an anti-incontinence procedure is being performed, should an inadvertent rectal injury occur. Once the sling incision is closed, a rectal injury can be repaired and the mesh is uninvolved.
6. **a. Traditional posterior colporrhaphy.** Both mesh-augmented repairs and levator plication report a higher incidence of complications, namely dyspareunia. As far as success is concerned, neither porcine nor biologic interposition grafts create a superior outcome when compared to traditional colporrhaphy.
7. **f. b and e.** Neither anterior colporrhaphy nor vaginal paravaginal repair have proven efficacy for the treatment of stress urinary incontinence. The other options are all viable treatment options.
8. **f. Anterior compartment prolapse repair also suspends the vaginal apex.** An anterior colporrhaphy restores the level 2 support, but has no effect on the apex, which is level 1 support. The other statements are true.
9. **d. Ba 0 with complaint of a vaginal bulge.** Only symptomatic prolapse requires repair. Repair of prolapse will have little effect on constipation, as this is usually a motility issue. Choices c and e demonstrate normal vaginal exams.
10. **d. Abdominal sacrohysteropexy with mesh.** Since she desires future fertility, we recommended that a hysterectomy not be performed for the management of her prolapse. This eliminates answers a, b, and e. Of the remaining two choices, the one that has the best potential of long-term success is the sacrohysteropexy. During this operation, a rectangular piece of mesh is sewn to the posterior vaginal wall (all the way down to the perineum), the cervix, and the lower uterine segment. An inverted U-shaped mesh is then sewn to the anterior vaginal wall all the way down to the pubic bone and lower portion of the anterior rectus abdominis

fascia. Due to the fact that mesh is fixated to the lower uterine segment, a C-section should be recommended as the mode of delivery.

11. **c. Manchester procedure.** The Manchester-Fothergill operation is the most appropriate transvaginal surgery for treatment of hypertrophic cervical elongation co-existing with a well-supported uterine corpus in a woman desiring uterine preservation.
12. **d. Le Forte colpocleisis.** The Leforte procedure is a partial colpocleisis in which matching rectangular sections of vaginal epithelium are excised from both the anterior and posterior vaginal walls and then fused together by sutures. The procedure can be performed under local anesthesia in patients with significant comorbidities.
13. **b. Sacrospinous cervicopexy.** Sacrospinous hysteropexy is a uterine-sparing procedure for cervical prolapse useful when an abdominal or laparoscopic/robotic procedure may be very difficult, due to presence of intraperitoneal adhesions or previous mesh-augmented ventral hernia repair.
14. **a. Uterosacral hysteropexy.** Uterosacral hysteropexy has been reported using both transvaginal and laparoscopic techniques in which the posterior cul de sac may be closed, the uterosacral ligaments are plicated in the midline, and if severed, then reattached to the cervix. This technique avoids the use of synthetic surgical mesh, for which this patient in the scenario has both known risk factors and complications from previous personal experience. This technique also reduces the risk of dyspareunia that can be associated with deviation of the vaginal axis related to unilateral sacrospinous fixation. The patient also has abdominal pathology and requests it be addressed at this surgery. An abdominal approach to her suspension procedure allows for evaluation and management of abdominal pathology.
15. **c. Sacrohysteropexy.** Sacrohysteropexy using a synthetic mesh bridge to suspend the posterior lower uterine segment/proximal cervix from the anterior sacrum (at S1-2) has been reported to be successful in a small series of women with collagen connective tissue disorders and in young women born with bladder exstrophy.

16. **e. Posterior colporrhaphy.** Posterior colporrhaphy is associated with the highest rate of dyspareunia—on average around 18% (0%–46%). In general, there is a high prevalence of dyspareunia in women with prolapse, but sexual function does improve with prolapse repair. Improvement in sexual function occurs more often in women who did not have a posterior repair. The uterosacral ligament suspension, anterior repair and enterocele repair are much less likely to cause dyspareunia. However, about 1% to 2% of patients who have undergone high uterosacral ligament suspension can develop sacral nerve entrapment (thigh, buttocks, and low back pain), which can affect sexual function (answer a, b, and d are incorrect). The TVT sling does not affect overall sexual function and satisfaction, but the partners of patients with mesh exposure (<3%) may complain of pain with intercourse (answer C is incorrect). The possible reasons for dyspareunia after posterior repair are the following:
    - Inadequate lateral dissection of the rectovaginal septum before plicating
    - Aggressive plication of lateral tissue, including levator muscles
    - Aggressive trimming of vaginal skin prior to closure
    - Poor surgical technique, making introitus too narrow
    - Development of vaginal scar tissue, infection, or hematoma
    - Vaginal atrophy
17. **c. Sciatic.** Prolonged hyperflexion and external rotation of the hips with knees extended can lead to sciatic nerve injury. This presents with foot drop, inability to flex the knee, and loss in sensation of the calf and lateral foot. One must be careful to properly position the patient; we recommend proper positioning be confirmed by the surgeon. Prolonged hip flexion, abduction, and external rotation may also lead to femoral nerve injury. This presents with weakness of the quadriceps muscle, which makes it difficult to extend the knee (i.e., knee gives out when trying to stand). There is also decrease in sensation of the anterior thigh and medial lower leg. The three big risk factors for neurologic injury because of patient

positioning are thinness (BMI <20 kg/m^2), diabetes mellitus, and time (each additional hour in lithotomy position increases risk).

18. **c. Abdominal supracervical hysterectomy with sacrocervicopexy ± anterior and posterior colporrhaphy.** The patient is sexually active, has no history of post-menopausal bleeding, and has a negative pap smear with no history of abnormal pap smears. A supracervical hysterectomy reduces the risk of graft extrusion at the time of the sacrocervicopexy. The procedure can be done open, laparoscopicly, or robotically; a, b, and d are incorrect because none of them address the apex. Answer e is incorrect because it is an obliterative procedure and she is currently sexually active.
19. **b. Uterosacral ligament suspension.** In reported studies uterosacral ligament suspension has a ureteral injury risk of 0% to 11% in prolapse surgery. In a systematic review, the rate of ureteral injury following transvaginal hysterectomy was 0.2 per 1000 surgeries. Ureteral injury can occur with anterior colporrhaphy but is uncommon.
20. **c. Osteomyelitis.** Sacral osteomyelitis is a well-known but fortunately rare complication of sacrocolpopexy, and common presenting symptoms include back pain, fevers, lethargy, and vaginal discharge. Sacral infections after sacrocolpopexies are likely to have a multifactorial origin. One hypothesis is that ischemia from devascularization during surgical dissection provides a microenvironment conducive to infection. It is conceivable that a suture placed too deeply into the vaginal canal or the sacral promontory provides a mechanism by which bacteria can be introduced into the abdominal cavity or into the periosteum of the sacral promontory. Similarly, placement of the sutures into the disc itself can lead to discitis. Sacrocolpopexy presacral sutures are often placed without knowledge of the location of the intervertebral discs. Herniated disc and spinal stenosis are clinical etiologies of back pain, but have not been reported as complications of mesh sacrocolpopexy. Though bowel obstruction can occur after

sacrocolpopexy, it would be unusual to present with back pain. This condition more commonly presents with nausea/vomiting.

21. **a. 1 cm lateral and slightly superior to the ischial spine.** In the area, there is decreased chance of injuring the pudendal nerves or vessels. The pudendal nerves and vessels lie directly posterior to the ischial spine. The suture should not be placed as close as these to the ischial spine. Superiorly and just behind the ligament lies the inferior gluteal vessels and hypogastric venous plexus.
22. **d. Colpocleisis.** Obliterative procedures have the lowest rates of prolapse recurrence and low rates of reoperation. When compared to native tissue repairs, the use of mesh in pelvic organ prolapse surgery is associated with increased rates of reoperations.

Chapter Review

1. The iliococcygeus and the coccygeus muscles fuse in the midline and attach to the coccyx, forming a complex called the *levator plate* that supports the upper vagina and cervix.
2. The urethra is fused to the anterior vaginal wall for much of its length.
3. Vaginal support is provided by the endopelvic connective tissues.
4. The cardinal and uterosacral ligaments provide level I support of the uterus and upper vagina; the endopelvic and pubocervical fascia provide level II support of the mid-vagina while it attaches to the arcus tendineus fasciae pelvis; the distal vagina attaches to the levator ani muscles and perineal body to provide for level III support.
5. Pelvic organ prolapse is for the most part a quality-of-life issue.
6. Anterior compartment defects can be central, lateral, or both.
7. Anterior colporrhaphy and paravaginal repairs are both ineffective alone in the treatment of stress urinary incontinence. They do not suspend the vaginal apex.
8. Sacrospinous ligament fixation may result in posterior displacement of the vaginal apex and increase the risk of anterior compartment prolapse.

9. If either the bladder or rectum is injured during pelvic organ prolapse repair, mesh should not be used.
10. Reoperation is needed for 29% of women who experience failed incontinence and prolapse surgery.
11. Synthetic mesh grafts erode in 10% of patients in whom they are used for the repair.
12. Sacrospinalis ligament fixation is an effective method of correcting vaginal apical prolapse. It results in posterior displacement of the vaginal apex.
13. Abdominal sacrocolpopexy maximizes functional vaginal length without significant distortion of the anatomic vaginal axis.
14. Colpocleisis obliterates a portion of the vagina and may be used to prevent vaginal vault prolapse. It precludes functional sexual activity.
15. Cystoscopy after the administration of indigo carmine or methylene blue is recommended as a routine practice after anterior colporrhaphy.
16. All women with advanced-stage anterior compartment prolapse should be screened for occult stress urinary incontinence.
17. The pain from sacrospinous ligament suspension is either gluteal or radiates down the leg posteriorly. It is neuropathic, and it occurs in 15% of patients.